

Course No.	Title of the Course	Course Structure	Prerequisite
	Digital Circuits and System	3L-0T-2P	—
COURSE OUTCOMES (COs) CO1: To get familiarized with number systems, codes, logic gates, logic families and Boolean algebra CO2: To analyze and understand the design process associated with combinational Circuits. CO3: To analyze and understand the design process associated with sequential circuit. CO4: To understand fundamental concepts of VHDL modeling for basic digital circuits. CO5: To develop basic understanding of programmable logic devices and convertors			
UNIT	CONTENTS		
1	Introduction to Digital Systems: Number Systems: Binary, octal and hexadecimal number systems, Number-Base Conversions, Complements of Numbers, Binary addition and subtraction (unsigned and one's and two's complement), Different codes and their properties (e.g., Gray, BCD, 84-2-1, 2421 etc.), Boolean equations and terminology (SOP, POS, minterm, maxterm), Logic implementations of equations (2-level, AND/OR, NAND/NOR, etc.), Boolean algebra axioms		
2	Combinational Logic: Karnaugh Maps up to 5-variables with don't care conditions for SOP and POS forms, Tabulation method, Half & Full Adder, Half & Full Subtractor, Parallel adders, BCD adder, Multiplexers, De-Multiplexers, Encoders, Decoders, Priority encoders, Magnitude Comparators, Code converters, ROM, PLA		
3	Sequential Logic: Latch, flip-flop, D, SR, JK, and T flip-flops (state table, excitation table, characteristic table), Race around condition, Master-Slave F/F, Direct inputs, conversion of one type of F/F to other, Registers, Counters (Synchronous and Asynchronous), Shift registers(PIPO, SISO, SIPO, PISO), Universal shift register, Ring counter & Johnson counter, Design & Analysis, Problem description in state table/graph, State assignment (binary, Gray), Next state and output equations. Mealy & Moore finite state machine		
4	Hardware Language: Verilog Introduction and its types of modeling style, Verilog implementation of adder, subtractor, multiplexer, decoders,		



	comparators etc, Verilog Implementation of Flip Flops, latches, Registers, Counters
5	Properties and characteristics (logic thresholds, delay, Noise margin, fan-in and fan-out, power dissipation etc.) of logic families, Transistor-level schematics of CMOS logic gates, Introduction to Digital to analog convertor (R-2R ladder network, Weighted ratio), Analog to digital convertor (Successive approximation, Flash type), Introduction to Programmable Logic Devices : SRAM, DRAM, CPLD and FPGA. Introduction to JTAG.

Books:

1. Digital design: With an Introduction to Verilog HDL: M. M. Mano & M. D. Ciletti, PHI
2. Digital principles & design : Donald D. Givone, McGraw Hill
3. Digital logic and computer design : M. M. Mano, PHI

CO → PO and CO → PSO Mapping:

CO/P O	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	1	2	1							1	3	2	3
CO2	3	3	1	2	2							1	3		3
CO3	3	3	1	3	2							1	3		3
CO4	3	3	1	3	2							1			3
CO5	3	3	1	2	1							1			3